

PRINCIPLES TO CHEMOTHERAPY

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INTRODUCTION TO CHEMOTHERAPY

2

- ▶ Chemotherapy is the form of treatment where drugs are used to eradicate pathogenic organisms or neoplastic cell in treatment of infectious diseases and cancer respectively
- ▶ One great principle on which chemotherapy is based is that of **selective toxicity**
- ▶ **Selective toxicity** entails that the chemotherapeutic drugs inhibits the vital function of invading microorganism or neoplastic cells
- ▶ The chemotherapeutic agent therefore, becomes toxic to the microorganism without harming the host
- ▶ This is dependent on the presence of the biochemical difference between the parasite and the host (qualitative and quantitative functional differences)

Classes of chemotherapeutic agents

3

The chemotherapeutic agents include the following;

i. Antimicrobial agents

- a). Antibacterial and antibiotics
- b). Antiviral drugs
- c). Anti-fungal drugs
- d). Anti parasitic drugs
 - Anti malaria drugs
 - Anti amoebic drugs
 - Anti helminthic drugs (Worm treatment)

ii. Antineoplastic drugs (Drugs used in cancer treatments)

ANTI-MICROBIAL DRUGS

- ▶ As one of the classes of chemotherapeutic agents, antimicrobial drugs are sub-classified as anti-bacterial, anti-fungal and anti-viral agents
- ▶ These agents include both natural ones called antibiotics and synthetic ones
- ▶ An antibiotic is a substance produced from or by one microbe which subsequently affects the growth or viability of another microbe

CLASSIFICATION OF ANTIMICROBIAL DRUGS

5

- ▶ Antimicrobial drugs are usually classified based on the following;
 - i. Site
 - ii. Mechanism of action
- ▶ Then they are further sub-classified based on the chemical structure
- ▶ Overall, these drugs include
 - i. Cell wall synthesis inhibitors
 - ii. Protein synthesis inhibitor
 - iii. Metabolic and nucleic acid synthesis inhibitors
 - iv. Cell membrane inhibitors

ANTIMICROBIAL ACTIVITY

6

The antimicrobial activity of a drug is characterized in the following way;

- i. Being either bactericidal or bacteriostatic against microorganism
- ii. Spectrum of activity against important groups of pathogens
- iii. Concentration time dependent effects on sensitive organisms

- ▶ **Bactericidal** drugs kill sensitive organism so as to rapidly lower the viable organisms after exposure to the drug
- ▶ **Bacteriostatic** drugs inhibits the growth but does not kill them and thus calling for the host immunological system to eliminate constant number of bacteria
- ▶ This also applies to drugs that which kill or inhibits the growth of fungi which are referred to as **fungicidal** and **fungistatic** drugs respectively

BACTERICIDAL VS BACTERIOSTATIC EFFECT

- Bactericidal drugs are mostly preferable when treating most bacteria infections
- This is because bactericidal typically produce a more rapid biological response, more clinical improvement and are less likely to elicit microbial resistance
- Bactericidal drugs have actions that induce lethal changes in microbial metabolism or block the activities that are essential for microbial viability (Here actually lies the basis for their mechanism of action) e.g. penicillins inhibiting bacterial cell wall synthesis
- In contrast, Bacteriostatic drugs usually inhibit the metabolic reaction that is needed for bacterial growth but is not necessary for survival e.g. sulphonamides inhibiting folate synthesis, an enzymatic co factor in synthesis of DNA components and amino acids

Bactericidal or bacteriostatic Cont'd

- ▶ Drugs like tetracyclines that reversibly inhibit bacterial protein synthesis are bacteriostatic
- ▶ Drugs that irreversibly inhibit bacterial protein synthesis like aminoglycosides are bactericidal
- ▶ Depending on the concentration of the drug and the bacterial species against which they are used, some drugs can either be bactericidal or bacteriostatic

ANTI-MICROBIAL SPECTRUM

9

Antimicrobial spectrum of activity is the primary determinant of the drug's clinical use

1. Narrow spectrum drugs antimicrobials

- ▶ These are antimicrobial which are active against a single species or a limited group of pathogens e.g. gram-positive bacteria

2. Broad spectrum antimicrobials

- ▶ These are antimicrobial drugs which are active against a wide range of pathogens

3. Extended spectrum antimicrobials

- ▶ These antimicrobial drugs have an effect on additional types of micro-organisms e.g. gram negative
- ▶ This results from the chemical modification of the drug structure intermediate range of activity

Considerations for Suitable use of a drug spectrum

- ▶ Narrow spectrum antimicrobial drugs are sometimes preferred because they target the specific pathogen without disturbing the normal flora of the gut or respiratory tract
- ▶ Broad spectrum antimicrobial drugs are sometimes preferred for initial treatment of an infection when the causative pathogen is not yet identified

CONCENTRATION AND TIME DEPENDENT EFFECTS

11

- ▶ Antimicrobial drugs exhibit various concentration time dependent effects which influence their clinical efficacy, dosage and frequency of administration

- ▶ Examples of concentration time dependent effects are
 - i. Minimum inhibitory concentration (**MIC**)
 - ❑ This is the minimum concentration of the drug that inhibits bacteria growth
 - ❑ Susceptibility or resistance of a particular strain of bacteria can be established based on the MIC

 - ii. Concentration dependent killing rate (**CDKR**)

 - iii. Post antibiotic effect (**PAE**)

Application of Concentration and time dependent effects

12

- ▶ Some aminoglycosides like tobramycin and some fluoroquinolones like ciprofloxacin exhibit a **Concentration dependent killing rate (CDKR)** against a large group of gram negative bacteria including *Pseudomonas aeruginosa* while Penicillins and other β -lactam antibiotics usually do not exhibit a CDKR
- ▶ After an antibacterial drug has been removed from the bacterial culture, evidence of a persistent effect on bacteria growth may exist, this effect being known as **Postantibiotic effect (PAE)** which many bactericidal antibiotics are known to exhibit against susceptible pathogens
- ▶ Penicillins show a PAE against gram positive cocci while aminoglycosides show PAE against gram negative bacilli
- ▶ Aminoglycosides exhibit both CDKR and PAE and hence the established treatment regimen where the entire daily dose is administered at one time which rapidly eliminates the bacteria while the PAE prevents any remaining bacteria from replicating for several hours after the drug has been eliminated from the body

MICROBIAL SENSITIVITY AND RESISTANCE

13

- ▶ Microbial sensitivity to antimicrobial drugs can be determined by various methods i.e. the broth dilution test (Test tube), the disk diffusion (Kirby – Bauer test) and the E – test method
- ▶ The broth and the E – test can be used to determine the MIC of a drug
- ▶ It is on the basis of MIC that the micro-organism can be classified as having, susceptibility, intermediate sensitivity or resistance to the tested drug
- ▶ These categories in terms of susceptibility are based on the relationship between the MIC and the peak serum concentration of the drug after administration of typical doses
- ▶ Generally, the peak serum concentration of the drug should be 4 – 10 times greater than the MIC in order for the pathogen to be susceptible
- ▶ It is worth noting that pathogens with intermediate sensitivity may respond to treatment with maximum doses of an antimicrobial agent

MICROBIAL RESISTANCE TO DRUGS

14

- ▶ Antimicrobial resistance (AMR) is the ability of a micro-organism to resist the effects of an antimicrobial medication that could previously treat the microbe successfully
- ▶ AMR has been proved a challenge to infectious disease management and has been documented to cause an estimated 700,000 deaths annually world wide (UN,2019)

Origin of resistance

- ▶ Resistance to antimicrobial agents can either be innate or acquired

Acquired resistance

This arises from the following;

- Mutation and selection
- Transfer of plasmids that confer drug resistance

NB: *It is the micro-organisms which become resistant to drugs and not drugs to micro-organisms*

Mutation and Selection

- ▶ Microbes can undergo spontaneous mutation to a form that is resistant to antimicrobial drugs
- ▶ These mutations occur at a constant rate per unit time i.e. if the organisms are exposed to antimicrobial drug the susceptible ones may be eradicated leaving the resistant mutant to multiply and become the dominant strain
- ▶ Mutations and selection of resistant mutant is increased by;
 - Exposure of a microorganism to sub-optimal drug concentrations
 - Prolonged exposure to the antimicrobial drug
- ▶ A successful outcome therefore, calls for the laboratory tests to guide the selection of antimicrobial drug while ensuring that the dosage and duration of therapy are adequate for the type of infection
- ▶ Additionally and whenever possible, bacteriological response to drug therapy should be verified by culturing samples of appropriate body fluids

Transferable resistance

16

- ▶ This usually results from bacterial conjugation and the transfer of plasmids(Extrachromosomal DNA) that confer drug resistance
- ▶ Bacterial conjugation enables a bacterium to donate a plasmid containing genes that encode proteins responsible for resistance to an antibiotic
- ▶ These genes are called **Resistance factors** which can either be transferred within the same species and between different species and thus mostly conferring what is termed **Multi-drug resistance**
- ▶ It is evidenced that the resident microflora (Normal flora) of human body can serve as reservoirs for resistant genes which they transfer to organisms that later invade or colonise the host
- ▶ Based on this transferable resistance principle, the various microorganism species need not to be present during the period in which the antibiotic is administered for resistance to ensue

Mechanisms of resistance

17

There are three major mechanisms by which resistance is actualized namely;

- i. Inactivation of drug by microorganism
- ii. Decreased accumulation of the drug by the microorganism
- iii. Reduced affinity for the drug by the target macromolecules

i. Inactivation of drug by microorganism

- ▶ This involves the production of enzymes by microorganism which destroy an important structural component of the drug e.g. beta lactamase destroying the lactam ring of penicillins
- ▶ This is an important mechanism of resistance to β -lactam antibiotics like penicillins and cephalosporins
- ▶ Resistance to aminoglycosides like gentamycin is partly caused by the production of drug inactivating enzymes that acetylate, adenylate or phosphorylates these antibiotics

ii. Decreased accumulation of an antibiotic by the microbe

- ▶ This occurs in two ways which is either by increased efflux (ejection or removal) or the decreased uptake of the drug
- ▶ Both these mechanisms contribute to the resistance of microbes to tetracyclines and fluoroquinolones
- ▶ Increased efflux of drug is mediated by membrane proteins that transport antimicrobial drugs out of the bacterial cell
- ▶ Decreased uptake of antimicrobial drugs can result from altered bacterial **porins** which are membrane proteins containing channels through which drugs and compounds enter bacteria
- ▶ Resistance to penicillins by gram negative bacilli is partly caused by altered porin channels that do not permit penicillin entry

iii. Reduced affinity of target molecule for antimicrobial drugs

- ▶ This is a common mechanism of microbial resistance to most classes of antibiotics
- ▶ This resistance often results from bacteria mutation followed by the selection of resistant mutants during exposure to antimicrobial drugs

Table showing mechanism of microbial resistance

20

MECHANISM	EXAMPLES
Inactivation of the drug by microbial enzymes	Inactivation of aminoglycosides by acetylase, adenylate synthetase and phosphorylase enzymes Inactivation of penicillins and other Beta lactam antibiotics by β -lactamase enzymes
Decreased accumulation of the drug by the microbe	Decreased uptake of β -lactam antibiotics due to altered porins in gram negative bacteria Decreased uptake and increased efflux of fluoroquinolones and tetracyclines
Reduced affinity of the target macromolecule for the drug	Reduced affinity for DNA gyrase for fluoroquinolones Reduced affinity of folate synthesis enzymes for sulphonamides and trimethoprim Reduced affinity of ribosomes for aminoglycosides, chloramphenicol, clindamycin, tetracycline, macrolides

SELECTION OF ANTIMICROBIAL DRUGS

21

Selection of an antimicrobial drug to be used for the treatment of a particular infection will significantly depend on the following:

1. Host factors

- ▶ The age of the patient
- ▶ Physiological status
- ▶ Immune competence of the patient

2. Drug factors

- ▶ Pharmacologic properties and antimicrobial activity
- ▶ Pharmacokinetic properties
- ▶ Adverse effect profile

3. Disease factors

- ▶ Causative of the infection
- ▶ Location and severity of infection

Host factors in drug selection

22

There are a number of the host factors that influence the choice of antimicrobial drugs as highlighted below;

- ▶ Most of the antimicrobials do cross the placenta e.g. tetracycline in pregnancy can lead to permanent discoloration of teeth of the child even as it is contraindicated in breastfeeding mothers and those below the age of twelve
- ▶ Many individuals are allergic to one or more antimicrobials, penicillins being the common cause of drug allergy
- ▶ The immune status is an important factor in determining the treatment success of which advanced age, diabetes, cancer chemotherapy and HIV are among the common causes of impaired immunity
- ▶ Therefore, patients with immune compromised state should be receive large doses and be on long duration of treatment
- ▶ Foreign bodies like the indwelling catheters may provide sites where microbes may be covered by a biofilm (Glycocalyx coating)
- ▶ Renal impairment may call for lower doses of antibiotics which renally excreted and similarly dose adjustment may be required in hepatic insufficiency e.g. chloramphenicol cannot be metabolized by neonates and hence the dosage per body weight should be lower than the adult ones

Antimicrobial activity

23

Antimicrobial agents can be selected based on the following;

- ▶ Laboratory tests in terms of culture, susceptibility and sensitivity tests
- ▶ Empiric selection – based on Knowledge of common organisms causing various infections and the preferred drugs for these organisms
- ▶ This Empiric therapy is used to treat serious infections until the results are available and is also used in treatment of minor upper respiratory and urinary tract infections based on the predictability of causative microorganism and their sensitivity to drugs, the situations in which the cost of microbial and drug sensitivity tests may not be justified
- ▶ Most infections are caused by a single microbial pathogen and thus may only need a single drug the approach called **monotherapy**
- ▶ The advantages of monotherapy are that it is less expensive, less toxic and has less effect on the host normal flora

Pharmacokinetic properties

24

- ▶ The pharmacokinetic properties that influence antibiotic selection include oral bioavailability, peak serum concentration, distribution to particular site of infection, routes of elimination and elimination half life
- ▶ An ideal antimicrobial drug for an ambulatory patient should have a good oral bioavailability with a long elimination half life e.g. Azithromycin
- ▶ Sites of infection that are not readily penetrated are the CNS, the prostate gland, the bone and the ocular tissues
- ▶ Penicillins can cross the blood brain barrier (BBB) when its inflamed while aminoglycosides do not whether the BBB is inflamed or not while chloramphenicol crosses the BBB with or without inflammation
- ▶ Because of the low pH of the prostatic fluid and the prostatic epithelium, some antimicrobial drugs are restricted entry into the prostate
- ▶ The route of elimination affects both the selection and use of antimicrobials i.e. fluoroquinolones which are renally excreted are more effective in treatment of urinary tract infections than the biliary metabolized or excreted drugs like erythromycin
- ▶ Drugs that are eliminated renally like aminoglycosides tends to accumulate in patients with renal impairment

Adverse effects profile

25

- ▶ All antimicrobial drugs can cause mild to severe adverse effects but the incidence and intensity varies among different classes of drugs
- ▶ It is therefore, important to weigh the **risk benefit ratio** when selecting drugs for treatment
- ▶ The β -lactam antibiotics like penicillins and cephalosporins cause a relatively low incidence of organ system toxicity and are often used in treatment of minor and common infections including those in pregnancy
- ▶ In contrast, aminoglycosides cause a relatively higher incidence and thus are reserved for treatment of more serious and life threatening conditions
- ▶ Fluoroquinolones and tetracyclines are intermediate in their adverse effect profile

COMBINATION DRUG THERAPY

26

Combination drug therapy are used in the following instances;

- ▶ Treatment of infections known or suspected to be caused by more than one pathogen (Mixed infections) e.g. intra-abdominal infections caused by both the aerobic and anaerobic micro-organisms from the GIT
- ▶ Treatment of life threatening infections like the hospital acquired (Nosocomial) infections like pneumonia till the causative organism is identified
- ▶ Treatment of infections where there is a likelihood of developing microbial resistance on monotherapy treatment e.g. T.B, HIV
- ▶ Administering antimicrobial in combination can result in any for the following; **antagonistic**, **additive**, **synergistic** or **indifferent** effects

Effects of Antimicrobial combination therapy

27

- ▶ **Antagonistic effect** – this is when the combined effect is less than the effect of either drug alone
- ▶ **Additive effect** – this is when the combined effect is the sum of the independent effects
- ▶ **Synergistic effect** - this is when the combined effect is greater than the sum of independent effects
- ▶ **Indifferent effects** – this is when the combined effect is similar to the greatest effect of either of the drugs
- ▶ Bacteriostatic drugs like chloramphenicol and tetracycline are antagonistic to bactericidal in that bactericidal drugs are usually more effective against rapidly dividing or growing bacteria
- ▶ Two bactericidal drugs targeting different bacteria function often exhibit additive or synergistic effects e.g. penicillins which inhibit cell wall synthesis often show additive or synergistic effects with aminoglycosides which inhibit protein synthesis

PROPHYLACTIC THERAPY

28

- ▶ The term prophylaxis means preventative or to prevent coming from the Greek word “Phylax,” meaning “to guard” and “Watching”
- ▶ Prophylactic treatment is frequently used in healthcare to minimize illness and disease
- ▶ Prevention of infections requires the sterilization of diagnostic and surgical tools, use of disinfectants to reduce environmental pathogens in hospitals and the disinfection of the skin and mucous membranes before invasive procedures
- ▶ Antimicrobial drugs are also administered to achieve prophylaxis in the following ways;
 - I. Prevent the occurrence of some infections in certain groups of patients e.g. immune compromised patients like HIV, Cancer etc
 - II. Reduce the incidence of infections associated with surgical and other invasive procedures
 - III. Prevent the disease transmission to close contacts of infected persons

Examples of prophylaxis treatment

1. Prevention of infection occurrence in immunocompromised patients

- ▶ Cotrimoxazole is used to prevent pneumocystis jiroveci pneumonia in HIV patients
- ▶ Isoniazid presumptive treatment is given to prevent T.B infection

2. Prevention of infection caused by invasive procedures

- ▶ Antibiotics are used to prevent endocarditis in persons with a history of valvular heart disease who are at risk of infection by viridans streptococci acquired during dental or oral procedures or in upper respiratory tract infections
- ▶ Antibiotics are routinely used to prevent tissue infection that can be acquired during a wide range of surgical procedures
- ▶ The selection of antimicrobials is dependent on the likely sources of bacteria, the skin being the major source of these bacteria during surgical procedures e.g. staphylococci

Examples of prophylaxis treatment cont'd

30

3. Prevention of Disease transmission

- ▶ Antimicrobial drugs are sometimes used to prevent the transmission of a highly contagious disease from an infected person or vector to an exposed person
- ▶ Antimicrobial drugs are also used to prevent malaria in persons who are travelling to regions where malaria is endemic
- ▶ Drugs are also used in preventing influenza type A in populations at increased risk for these diseases

Assignment

Question:

Discuss antimicrobial stewardship, stating its significance, the different approaches and the role of a biomedical scientists

- ▶ Due date:
- ▶ It should be typed, not less than 3 pages and not more than 5 pages excluding cover and reference pages
- ▶ Submission method: via email - lundsb82@gmail.com

END